

PH102

After Mid Semester Exams

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Chapter-9

Electromagnetic Waves in Vacuum

Generalization for Propagation of Monochromatic Plane EM Waves in an Arbitrary Direction

Obviously, there is nothing special/profound with regard to plane *EM* waves propagating in a specific direction in free space/the vacuum. They can propagate in any direction. We can easily generalize the mathematical description for monochromatic plane *EM* waves traveling in an arbitrary direction as follows:

$$\vec{\tilde{E}}(\mathbf{r}, t) = \tilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{\mathbf{n}}$$

$$\vec{\tilde{B}}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{k}} \times \vec{E}(\mathbf{r}, t) = \frac{1}{c} \tilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} (\vec{k} \times \hat{\mathbf{n}})$$

We must have $\vec{\tilde{B}}(\mathbf{r}, t) \perp \vec{\tilde{E}}(\mathbf{r}, t) \perp \hat{k}$

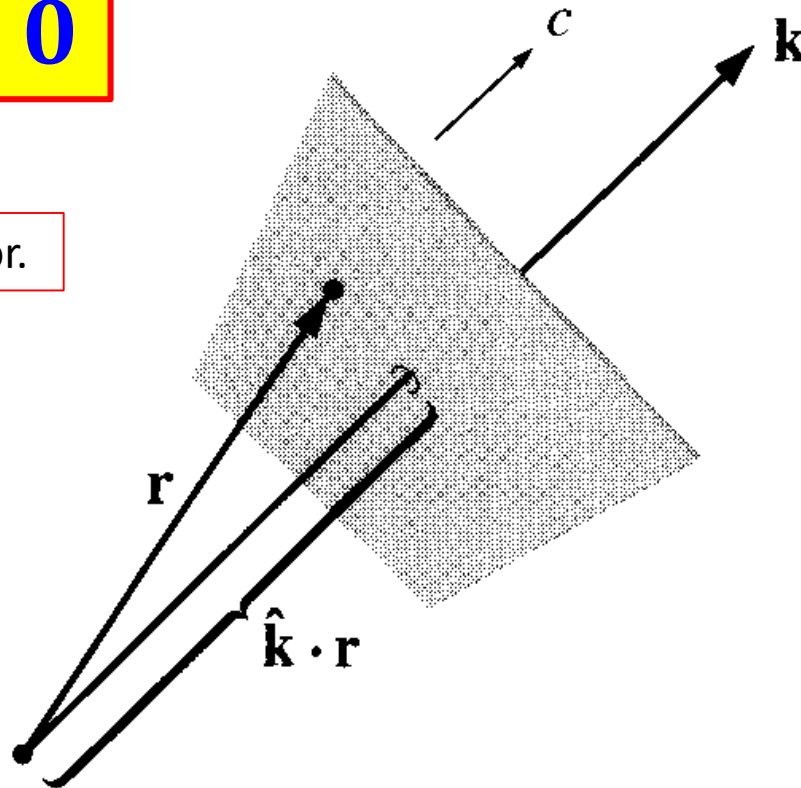
$$\vec{\tilde{E}} \cdot \vec{\tilde{B}} = 0$$

$$\vec{\tilde{E}} \cdot \hat{k} = 0$$

$$\vec{\tilde{B}} \cdot \hat{k} = 0$$

$$\hat{n} \cdot \hat{k} = 0$$

is the polarization vector.



The Direction of Propagation of a Monochromatic Plane EM Wave : $\hat{\mathbf{k}}$

The Real/Physical (Instantaneous) *EM* Fields are:

$$\vec{E}(r, t) = \text{Re} \left(\vec{\tilde{E}}(r, t) \right) = E_0 \cos(\vec{\mathbf{k}} \cdot \vec{\mathbf{r}} - \omega t + \delta) \hat{\mathbf{n}}$$

$\hat{\mathbf{n}} = \text{Polarization Vector } (\parallel \vec{E})$

$$\vec{B}(r, t) = \text{Re} \left(\vec{\tilde{B}}(r, t) \right) = B_0 \cos(\vec{\mathbf{k}} \cdot \vec{\mathbf{r}} - \omega t + \delta) (\hat{\mathbf{k}} \times \hat{\mathbf{n}})$$

$$B_0 = \frac{E_0}{c} \quad \text{In free space}$$

$$\vec{B}(r, t) = \text{Re} \left(\vec{\tilde{B}}(r, t) \right) = \frac{E_0}{c} \cos(\vec{\mathbf{k}} \cdot \vec{\mathbf{r}} - \omega t + \delta) (\hat{\mathbf{k}} \times \hat{\mathbf{n}})$$

Energy and Momentum in Electromagnetic Waves

The energy per unit volume stored in electromagnetic fields is

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$$

In the case of a monochromatic plane wave

$$B^2 = \frac{1}{c^2} E^2 \qquad B^2 = \mu_0 \epsilon_0 E^2$$

So the electric and magnetic contributions are equal:

$$u = \epsilon_0 E^2 = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)$$

As the wave travels, it carries this energy along with it. The energy flux density (energy per unit area, per unit time) transported by the fields is given by the Poynting vector

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

For monochromatic plane waves propagating in the z direction,

$$\vec{E}(r, t) = E_0 \cos(kz - \omega t + \delta) \hat{x}$$

$$\vec{B}(r, t) = \frac{E_0}{c} \cos(kz - \omega t + \delta) \hat{y}$$

$$u = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)$$

$$\vec{S} = c \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta) \hat{z} = c u \hat{z}$$

All we want is the average value

Now, the average of cosine-squared over a complete cycle is $\frac{1}{2}$
so

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2$$

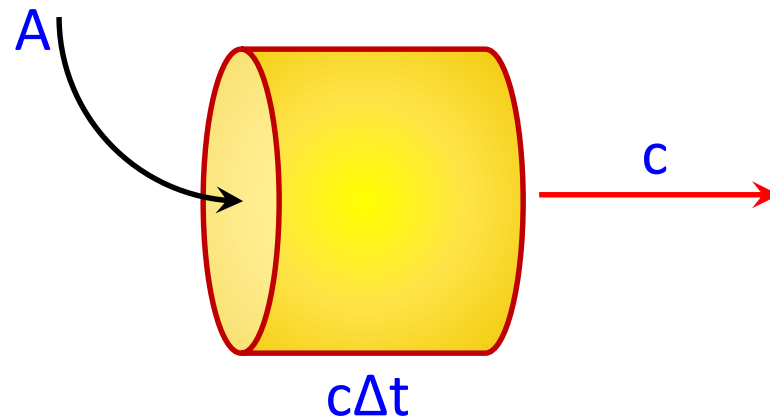
$$\langle s \rangle = \frac{1}{2} c \epsilon_0 E_0^2 \hat{z}$$

Here the brackets, $\langle \quad \rangle$, to denote the (time) average over a complete cycle

The average power per unit area transported by an electromagnetic wave is called the intensity:

$$I = \langle s \rangle = \frac{1}{2} c \epsilon_0 E_0^2 \hat{z}$$

Notice that S is the energy density (u) times the velocity of the waves ($c\hat{z}$)-as it should be. For in a time Δt , a length $c\Delta t$ passes through area A (Figure.), carrying with it an energy $u A c\Delta t$.



Example 1:

Write down the (real) electric and magnetic fields for a monochromatic plane wave of amplitude E_0 , frequency ω , and phase angle zero that is

- (a) traveling in the negative x direction and polarized in the z direction;
- (b) traveling in the direction from the origin to the point (1,1,1), with polarization parallel to the x-z plane. In each case, sketch the wave, and give the explicit Cartesian components of $\vec{k}$ and $\hat{n}$.

Solution :

(a) Monochromatic plane wave traveling in the negative 'x' direction and polarized in the 'z' direction

Propagation vector $\vec{k} = -\frac{\omega}{c}\hat{x}$ and polarization vector $\hat{n} = \hat{z}$

$$\vec{k} \cdot \vec{r} = -\frac{\omega}{c}\hat{x} \cdot (x\hat{x} + y\hat{y} + z\hat{z})$$

$\Rightarrow$

$$\vec{k} \cdot \vec{r} = -\frac{\omega}{c}x$$

Electric field

$$\tilde{E}(r, t) = \tilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{n}$$

Real component of Electric field is

$$\tilde{E}(r, t) = \widetilde{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \hat{n}$$

$$\mathbf{E}(x, t) = E_0 \cos \left(-\frac{\omega}{c} x - \omega t \right) \hat{z}$$

$$\mathbf{E}(x, t) = E_0 \cos \left(\frac{\omega}{c} x + \omega t \right) \hat{z}$$

Magnetic field

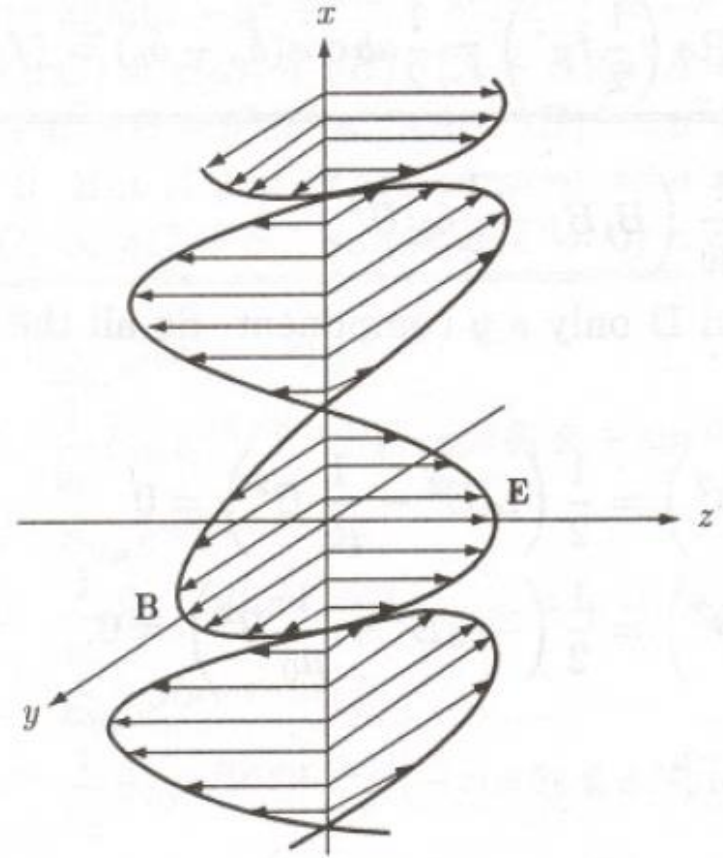
$$\mathbf{B} = \frac{\hat{k} \times \mathbf{E}}{c}$$

Where $\hat{k} = -\hat{x}$



Real component of magnetic field is

$$\mathbf{B}(x, t) = \frac{E_0}{c} \cos \left(\frac{\omega}{c} x + \omega t \right) \hat{y}$$



(b) Monochromatic plane wave traveling in the direction from the origin to the point (1,1,1)

$$\vec{k} = \frac{\omega}{c} \left(\frac{\hat{x} + \hat{y} + \hat{z}}{\sqrt{3}} \right)$$

polarization vector $\hat{n}$ is parallel to the x-z plane ,
it must have the form $\alpha\hat{x} + \beta\hat{z}$

Since $\hat{n} \cdot \vec{k} = 0 \Rightarrow \alpha = -\beta$

As it is a unit vector $\alpha = \frac{1}{\sqrt{2}}$

Therefore $\hat{n} = \left(\frac{\hat{x} - \hat{z}}{\sqrt{2}} \right)$

$$\vec{k} \cdot \vec{r} = \frac{\omega}{c} \left(\frac{\hat{x} + \hat{y} + \hat{z}}{\sqrt{3}} \right) \cdot (x\hat{x} + y\hat{y} + z\hat{z})$$

$$\tilde{E}(r, t) = \widetilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{n}$$

$$\Rightarrow \vec{k} \cdot \vec{r} = \frac{\omega}{\sqrt{3}c} (x + y + z)$$

Electric field

$$\tilde{E}(r, t) = \widetilde{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \hat{n}$$

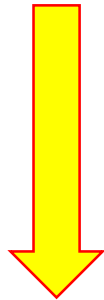
Real component of Electric field is

$$\mathbf{E}(x, t) = E_0 \cos \left(\frac{\omega}{\sqrt{3}c} (x + y + z) - \omega t \right) \left(\frac{\hat{x} - \hat{z}}{\sqrt{2}} \right)$$

Magnetic field

$$\mathbf{B} = \frac{\hat{k} \times \mathbf{E}}{c}$$

Where $\hat{k} = \left(\frac{\hat{x} + \hat{y} + \hat{z}}{\sqrt{3}} \right)$

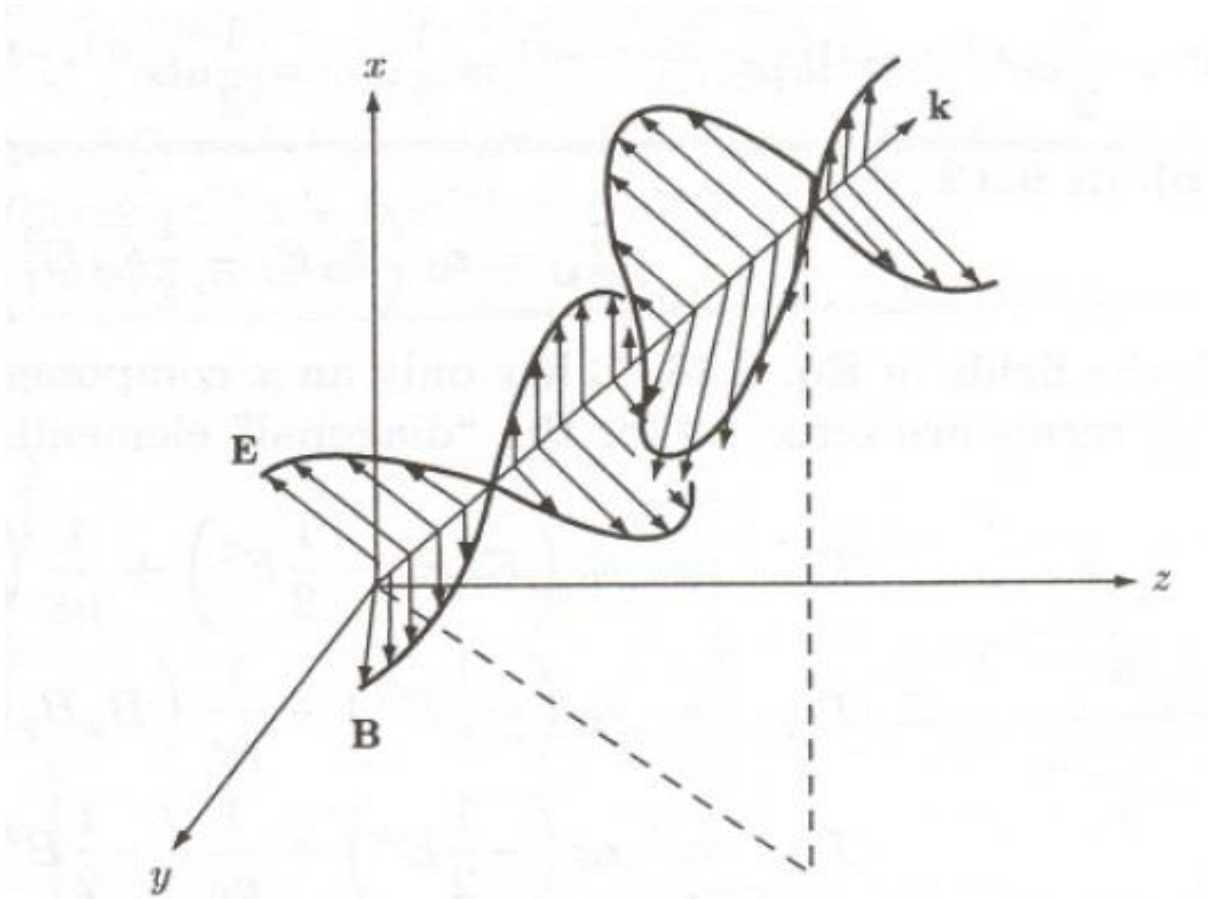


$$\hat{k} \times \hat{n} = \frac{-\hat{x} + 2\hat{y} - \hat{z}}{\sqrt{6}}$$

Real component of magnetic field is

$$\mathbf{B}(x, t) = \frac{E_0}{c} \cos \left(\frac{\omega}{\sqrt{3}c} (x + y + z) - \omega t \right) \left(\frac{-\hat{x} + 2\hat{y} - \hat{z}}{\sqrt{6}} \right)$$

So, direction of electric field is along $\hat{n} = \left(\frac{\hat{x} - \hat{z}}{\sqrt{2}} \right)$ direction and direction of magnetic field is along $\frac{-\hat{x} + 2\hat{y} - \hat{z}}{\sqrt{6}}$ direction .



Example 2:

In the complex notation there is a clever device for finding the time average of a product. Suppose $f(\mathbf{r}, t) = a \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_a)$ and $g(\mathbf{r}, t) = b \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_b)$. Show that $\langle fg \rangle = \frac{1}{2} \text{Re}(\tilde{f} \tilde{g}^*)$ where the star denotes complex conjugation. [Note that this only works if the two waves have the same $\mathbf{k}$ and ω , but they need not have the same amplitude or phase.]

Solution :

Given

$$f(r, t) = a \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_a)$$

$$g(r, t) = b \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_b)$$

$$\langle fg \rangle = \frac{1}{T} \int_0^T [a \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_a)] [b \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_b)] dt$$

$$\langle fg \rangle = \frac{ab}{2T} \int_0^T 2 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_a) \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta_b) dt$$

$$2 \cos(A) \cos(B) = \cos(A + B) + \cos(A - B)$$

$$\langle fg \rangle = \frac{ab}{2T} \int_0^T [\cos(2\mathbf{k} \cdot \mathbf{r} - 2\omega t + \delta_a + \delta_b) + \cos(\delta_a - \delta_b)] dt$$

$$\langle fg \rangle = \frac{ab}{2T} \cos(\delta_a - \delta_b) T$$

$$\langle fg \rangle = \frac{1}{2} a b \cos(\delta_a - \delta_b)$$

In the complex notation $\tilde{f} = \tilde{a} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$ $\tilde{g} = \tilde{b} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$

Where $\tilde{a} = a e^{i\delta_a}$ and $\tilde{b} = b e^{i\delta_b}$

So,

$$\frac{1}{2}(\tilde{f}\tilde{g}^*) = \frac{1}{2}\tilde{a}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\tilde{b}^*e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$$

$$\frac{1}{2}(\tilde{f}\tilde{g}^*) = \frac{1}{2}\tilde{a}\tilde{b}^*$$

$$\frac{1}{2}(\tilde{f}\tilde{g}^*) = \frac{1}{2}a b e^{i(\delta_a-\delta_b)}$$

Therefore

$$\frac{1}{2}\text{Re}(\tilde{f}\tilde{g}^*) = \frac{1}{2}a b \cos(\delta_a - \delta_b) = \langle fg \rangle$$

Example 3:

The intensity of sunlight hitting the earth is about 1300 W/m^2 . If sunlight strikes a perfect absorber, what pressure does it exert? How about a perfect reflector? What fraction of atmospheric pressure does this amount to?

Solution :

Radiation pressure

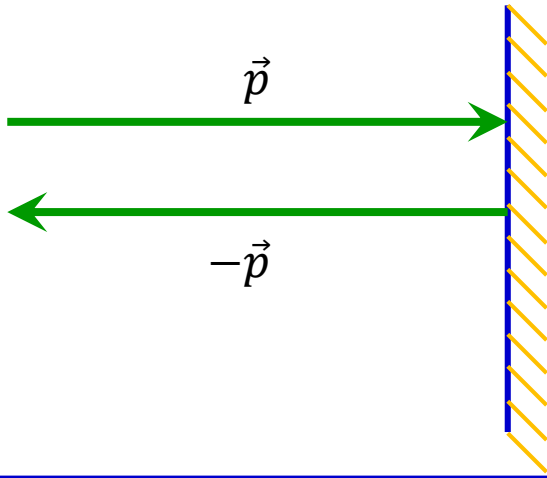
$$P = \frac{I}{c}$$
$$= \frac{1300 \text{ W/m}^2}{3 \times 10^8 \text{ m/sec}}$$

$$P = 4.3 \times 10^{-6} \text{ N/m}^2$$

For a perfect reflector the pressure is twice

$$P' = 8.6 \times 10^{-6} \text{ N/m}^2$$

For Perfect Reflector



$$\Delta\vec{p}_{abs} = \vec{p}_{inc} - \vec{p}_{ref} = \vec{p} - (0) = \vec{p}$$

$$\Delta\vec{p}_{ref} = \vec{p}_{inc} - \vec{p}_{ref} = \vec{p} - (-\vec{p}) = 2\vec{p}$$

$$P_{abs} = \frac{\vec{F}}{A} = \frac{1}{A} \frac{\Delta\vec{p}_{abs}}{\Delta t} = \frac{1}{A} \frac{\vec{p}}{\Delta t}$$

$$P_{ref} = \frac{\vec{F}}{A} = \frac{1}{A} \frac{\Delta\vec{p}_{ref}}{\Delta t} = \frac{1}{A} \frac{2\vec{p}}{\Delta t} = 2 P_{abs}$$

For a perfect reflector the pressure is twice

$$P' = 8.6 \times 10^{-6} \text{ N/m}^2$$

Atomic pressure is $1.03 \times 10^5 \text{ N/m}^2$

So , fraction of atmospheric pressure amount to pressure of light on a reflector is

$$= \frac{8.6 \times 10^{-6}}{1.03 \times 10^5}$$

$$= 8.3 \times 10^{-11} \text{ atmospheres}$$

Example 4:

Show that the standing wave $f(z, t) = A \sin(kz) \cos(kvt)$ satisfies the wave equation, and express it as the sum of a wave traveling to the left and a wave traveling to the right .

Solution :

Given , $f(z, t) = A \sin(kz) \cos(kvt)$ ----- (a)

$$\frac{\partial f}{\partial z} = Ak \cos(kz) \cos(kvt)$$

$$\frac{\partial^2 f}{\partial z^2} = -Ak^2 \sin(kz) \cos(kvt)$$

$$\frac{\partial f}{\partial t} = -A(vk) \sin(kz) \sin(kvt)$$

$$\frac{\partial^2 f}{\partial t^2} = -A(vk)^2 \sin(kz) \cos(kvt)$$



$$\frac{\partial^2 f}{\partial t^2} = v^2 \frac{\partial^2 f}{\partial z^2}$$

Therefore equation (a) satisfy the wave equation .

$$f(z, t) = A \sin(kz) \cos(kvt)$$

$$f(z, t) = \frac{A}{2} [2 \sin(kz) \cos(kvt)]$$

$2 \sin(A) \cos(B) = \sin(A - B) + \sin(A + B)$

$$f(z, t) = \frac{A}{2} [\sin(kz - kvt) + \sin(kz + kvt)] \quad \text{----- (b)}$$

General solution of wave equation is

$$f(z, t) = g(z - vt) + h(z + vt)$$

$$\begin{aligned} \Rightarrow g(z - vt) &= \frac{A}{2} \sin k(z - vt) \\ \Rightarrow h(z + vt) &= \frac{A}{2} \cos k(z + vt) \end{aligned} \quad \left. \vphantom{\begin{aligned} \Rightarrow g(z - vt) &= \frac{A}{2} \sin k(z - vt) \\ \Rightarrow h(z + vt) &= \frac{A}{2} \cos k(z + vt) \end{aligned}} \right\} \text{On comparing with eq. (b)}$$

So equation (a) can be expressed as the sum of a wave traveling to the left and a wave traveling to the right .

Generalization for Propagation of Monochromatic Plane EM Waves in an Arbitrary Direction

Obviously, there is nothing special/profound with regard to plane *EM* waves propagating in a specific direction in free space/the vacuum. They can propagate in any direction. We can easily generalize the mathematical description for monochromatic plane *EM* waves traveling in an arbitrary direction as follows:

One Dimensional

$$\text{If } \vec{k} = k\hat{z}$$

$$|\vec{k}| = k \quad \vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\vec{k} \cdot \vec{r} = k\hat{z} \cdot (x\hat{x} + y\hat{y} + z\hat{z}) = kz$$

Three-Dimensional

$$\vec{k} = k_x \hat{x} + k_y \hat{y} + k_z \hat{z}$$

$$|\vec{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2}$$

$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\begin{aligned}\vec{k} \cdot \vec{r} &= (k_x \hat{x} + k_y \hat{y} + k_z \hat{z}) \cdot (x \hat{x} + y \hat{y} + z \hat{z}) \\ &= k_x x + k_y y + k_z z\end{aligned}$$

$$k_x = k \cos \Theta_x$$

$$k_y = k \cos \Theta_y$$

$$k_z = k \cos \Theta_z$$

Where Θ_x , Θ_y , and Θ_z are direction cosines w.r.t. $\hat{x}$, $\hat{y}$ and $\hat{z}$

Direction Co-sines in spherical co-ordinates

$$\cos \Theta_x = \hat{k} \cdot \hat{x} = \sin \theta \cos \varphi$$

$$\cos \Theta_y = \hat{k} \cdot \hat{y} = \sin \theta \sin \varphi$$

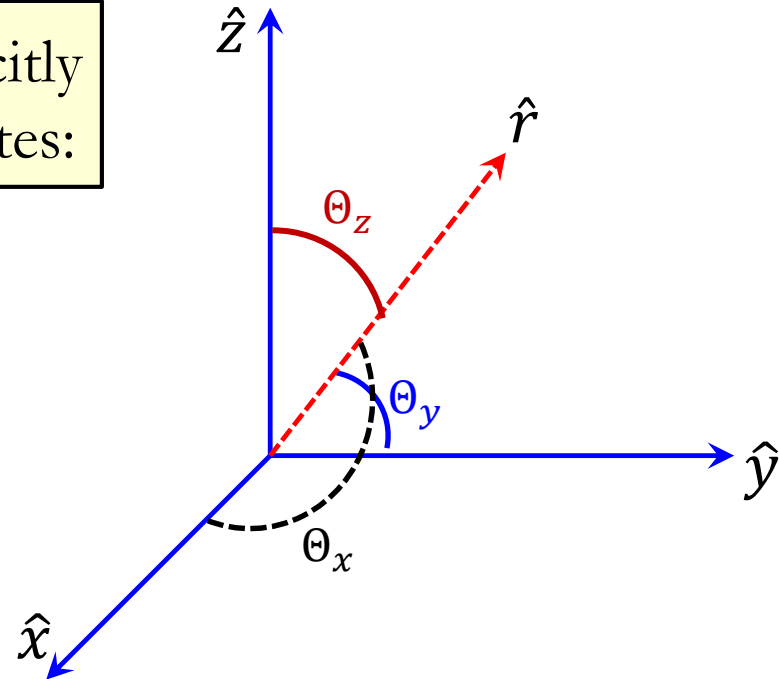
$$\cos \Theta_z = \hat{k} \cdot \hat{z} = \cos \theta$$

If *e.g.* $\vec{k} \parallel \vec{r}$ then: $\vec{k} \cdot \vec{r} = kr$. We explicitly demonstrate this in spherical polar coordinates:

$$k_x = k \cos \Theta_x = k \sin \theta \cos \varphi$$

$$k_y = k \cos \Theta_y = k \sin \theta \sin \varphi$$

$$k_z = k \cos \Theta_z = k \cos \theta$$



$$x = r \cos \Theta_x = r \sin \theta \cos \varphi$$

$$y = r \cos \Theta_y = r \sin \theta \sin \varphi$$

$$z = r \cos \Theta_z = r \cos \theta$$

Then

$$\vec{k} \cdot \vec{r} = k_x x + k_y y + k_z z = kx \cos \Theta_x + ky \cos \Theta_y + kz \cos \Theta_z$$

$$\vec{k} \cdot \vec{r} = kr \cos^2 \Theta_x + kr \cos^2 \Theta_y + kr \cos^2 \Theta_z$$

$$\vec{k} \cdot \vec{r} = kr \sin^2 \theta \cos^2 \varphi + kr \sin^2 \theta \sin^2 \varphi + kr \cos^2 \theta$$

$$\vec{k} \cdot \vec{r} = kr(\sin^2 \theta \cos^2 \varphi + \sin^2 \theta \sin^2 \varphi + \cos^2 \theta)$$

$$\vec{k} \cdot \vec{r} = kr(\sin^2 \theta (\underbrace{\cos^2 \varphi + \sin^2 \varphi}_{1}) + \cos^2 \theta)$$

$$\vec{k} \cdot \vec{r} = kr(\underbrace{\sin^2 \theta + \cos^2 \theta}_{1})$$

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$$\vec{k} \cdot \vec{r} = kr$$

Thus, most generally, we can write the $\vec{E}(\mathbf{r}, t)$ and $\vec{B}(\mathbf{r}, t)$ fields as:

$$\vec{E}(\mathbf{r}, t) = \tilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{\mathbf{n}}$$

Where, $\hat{\mathbf{n}}$ =Polarization vector; $\hat{\mathbf{n}} \perp \hat{\mathbf{k}}$; i.e. $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = 0$ because $\mathbf{E}$ is transverse"

$$\vec{B}(\mathbf{r}, t) = \tilde{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} (\hat{\mathbf{k}} \times \hat{\mathbf{n}})$$

$$\vec{B}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{k}} \times \vec{E}(\mathbf{r}, t) = \frac{1}{c} \tilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} (\vec{k} \times \hat{\mathbf{n}})$$

$$\vec{B}(\mathbf{r}, t) = \tilde{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} (\vec{k} \times \hat{\mathbf{n}})$$

$$\vec{\widetilde{E}}(r, t) = \widetilde{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \hat{\mathbf{n}}$$

$$\vec{\widetilde{B}}(r, t) = \widetilde{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} (\vec{k} \times \hat{n})$$

We must have $\vec{\widetilde{B}}(r, t) \perp \vec{\widetilde{E}}(r, t) \perp \hat{k}$

$$\vec{\widetilde{E}} \cdot \vec{\widetilde{B}} = 0$$

$$\vec{\widetilde{E}} \cdot \hat{k} = 0$$

$$\vec{\widetilde{B}} \cdot \hat{k} = 0$$

The Direction of Propagation of a Monochromatic Plane EM Wave : $\hat{\mathbf{k}}$

The Real/Physical (Instantaneous) *EM* Fields are:

$$\vec{E}(r, t) = \text{Re} \left(\vec{\tilde{E}}(r, t) \right) = E_0 \cos(\vec{\mathbf{k}} \cdot \vec{r} - \omega t + \delta) \hat{\mathbf{n}}$$

$\hat{\mathbf{n}}$ = **Polarization Vector** ($\parallel \vec{E}$)

$$\vec{B}(r, t) = \text{Re} \left(\vec{\tilde{B}}(r, t) \right) = B_0 \cos(\vec{\mathbf{k}} \cdot \vec{r} - \omega t + \delta) (\hat{\mathbf{k}} \times \hat{\mathbf{n}})$$

$$B_0 = \frac{E_0}{c} \quad \text{In free space}$$

Energy and Momentum in Electromagnetic Waves

The energy per unit volume stored in electromagnetic fields is

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$$

In the case of a monochromatic plane wave

$$B^2 = \frac{1}{c^2} E^2 \qquad B^2 = \mu_0 \epsilon_0 E^2$$

So the electric and magnetic contributions are equal:

$$u = \epsilon_0 E^2 = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)$$

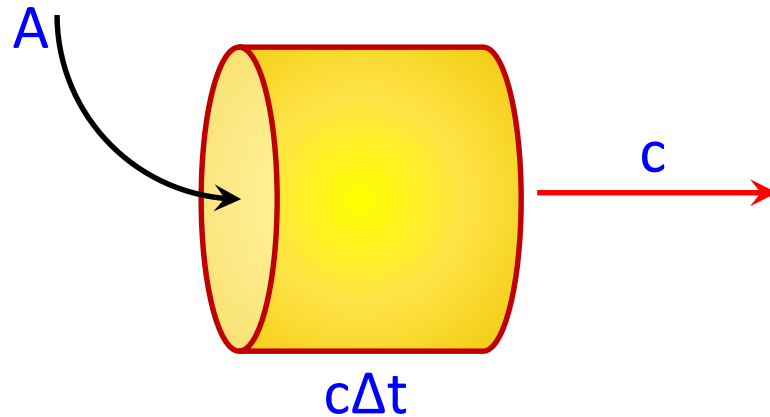
As the wave travels, it carries this energy along with it. The energy flux density (energy per unit area, per unit time) transported by the fields is given by the Poynting vector

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

For monochromatic plane waves propagating in the z direction,

$$\vec{S} = c\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta) \hat{z}$$

Notice that S is the energy density (u) times the velocity of the waves ($c\hat{z}$)-as it should be. For in a time Δt , a length $c \Delta t$ passes through area A (Figure.), carrying with it an energy $uAc \Delta t$.



The energy per unit time, per unit area, transported by the wave is therefore $u c$.

Electromagnetic fields not only carry energy, they also carry momentum.

The momentum density stored in the fields

$$\rho = \frac{1}{c} u \hat{z}$$

In the case of light, the wavelength is so short (5×10^{-7} m), and the period so brief (10^{-15} s), that any macroscopic measurement will encompass many cycles.

All we want is the average value

Now, the average of cosine-squared over a complete cycle is $\frac{1}{2}$
so

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2$$

$$\langle s \rangle = \frac{1}{2} c \epsilon_0 E_0^2 \hat{z}$$

$$\langle q \rangle = \frac{1}{2c} \epsilon_0 E_0^2 \hat{z}$$

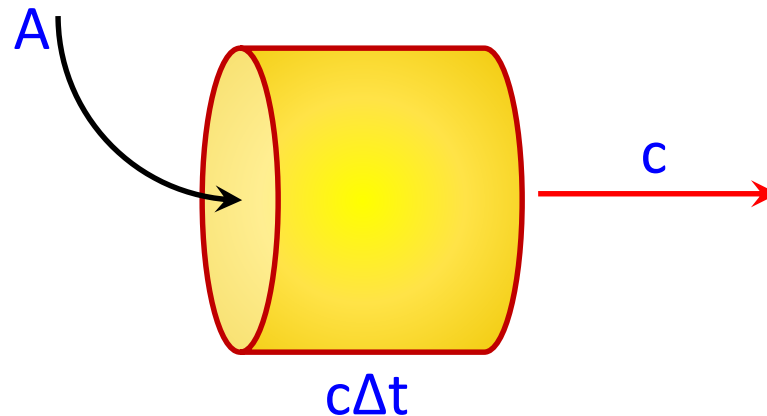
Here the brackets, $\langle \quad \rangle$, to denote the (time) average over a complete cycle

The average power per unit area transported by an electromagnetic wave is called the intensity:

$$I = \langle s \rangle = \frac{1}{2} c \epsilon_0 E_0^2 \hat{z}$$

When light falls on a perfect absorber it delivers its momentum to the surface. In a time Δt the momentum transfer is (Figure) $\Delta p = \langle q \rangle A c \Delta t$, so the radiation pressure (**average force per unit area**) is

$$P = \frac{1}{A} \frac{\Delta p}{\Delta t}$$



$$P = \frac{1}{A} \frac{\Delta p}{\Delta t} \quad \leftarrow \Delta p = \langle q \rangle A c \Delta t$$

$$P = \frac{1}{A} \frac{\langle q \rangle A c \Delta t}{\Delta t}$$

$$P = \langle q \rangle c \Delta t / \Delta t$$

$$\langle q \rangle = \frac{1}{2c} \epsilon_0 E_0^2 \hat{z}$$

$$P = \frac{1}{2c} \epsilon_0 E_0^2 c \hat{z}$$

$$P = \frac{1}{2} \epsilon_0 E_0^2 \hat{z}$$

Radiation pressure

$$P = \frac{1}{2} \epsilon_0 E_0^2 \hat{z} \quad \leftarrow \quad I = \frac{1}{2} c \epsilon_0 E_0^2 \hat{z}$$

Radiation pressure

$$P = \frac{I}{c}$$

Example (9.32 Griff):

The "inversion theorem" for Fourier transforms states that

$\tilde{\phi}(z) = \int_{-\infty}^{\infty} \tilde{\phi}(k) e^{ikz} dk \Leftrightarrow \tilde{\phi}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{\phi}(z) e^{-ikz} dz$. Use this to determine $\tilde{A}(k)$ in terms of $f(z, 0)$ and $\dot{f}(z, 0)$

Solution :

Griffith page # 369
Equation (9.20)

Expression of sinusoidal waves

$$\tilde{f}(z, t) = \int_{-\infty}^{+\infty} \tilde{A}(k) e^{i(kz - \omega t)} dk$$

$$\tilde{f}(z, 0) = \int_{-\infty}^{\infty} \tilde{A}(k) e^{ikz} dk \Leftrightarrow \tilde{f}^*(z, 0) = \int_{-\infty}^{\infty} \tilde{A}^*(k) e^{-ikz} dk$$

Let $l = -k$

$$\tilde{f}^*(z, 0) = \int_{\infty}^{-\infty} \tilde{A}^*(-l) e^{ilz} (-dl)$$

$$\tilde{f}^*(z, 0) = \int_{-\infty}^{\infty} \tilde{A}^*(-l) e^{ilz} dl$$

$$= \int_{-\infty}^{\infty} \tilde{A}^*(-k) e^{ikz} dk$$

(renaming the dummy variable $l \rightarrow k$).

$$f(z, 0) = \operatorname{Re}[\tilde{f}(z, 0)] = \frac{1}{2} [\tilde{f}(z, 0) + \tilde{f}^*(z, 0)]$$

$$\tilde{f}(z, 0) = \int_{-\infty}^{+\infty} \frac{1}{2} [\tilde{A}(k) + \tilde{A}^*(-k)] e^{ikz} dk$$

So

$$\frac{1}{2} [\tilde{A}(k) + \tilde{A}^*(-k)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(z, 0) e^{-ikz} dz \quad \text{-----} \quad (i)$$

Now

$$\dot{\tilde{f}}(z, t) = \int_{-\infty}^{+\infty} \tilde{A}(k) (-i\omega) e^{i(kz - \omega t)} dk$$

$$\Rightarrow \dot{\tilde{f}}(z, 0) = \int_{-\infty}^{+\infty} \tilde{A}(k) (-i\omega) e^{ikz} dk$$

Note that , $\omega = |k|v = \text{function of } k$

$$\dot{\tilde{f}}^*(z, 0) = \int_{-\infty}^{+\infty} [i\omega \tilde{A}^*(k)] e^{-ikz} dk$$

$$\dot{\tilde{f}}^*(z, 0) = \int_{-\infty}^{+\infty} [i|k|v \tilde{A}^*(k)] e^{-ikz} dk$$

Let $l = -k$

$$\dot{\tilde{f}}^*(z, 0) = \int_{\infty}^{-\infty} [i|l|v \tilde{A}^*(-l)] e^{ilz} (-dl)$$

renaming the dummy variable $l \rightarrow k$

$$\dot{\tilde{f}}^*(z, 0) = \int_{-\infty}^{\infty} [i|k|v\tilde{A}^*(-k)] e^{ikz} dk$$

$$\dot{\tilde{f}}^*(z, 0) = \int_{-\infty}^{\infty} [i\omega\tilde{A}^*(-k)] e^{ikz} dk$$

Therefore

$$\dot{f}(z, 0) = \text{Re} [\dot{\tilde{f}}(z, 0)] = \frac{1}{2} [\dot{\tilde{f}}(z, 0) + \dot{\tilde{f}}^*(z, 0)]$$

$$\dot{\tilde{f}}(z, 0) = \int_{-\infty}^{+\infty} \frac{1}{2} [-i\omega\tilde{A}(k) + i\omega\tilde{A}^*(-k)] e^{ikz} dk$$

So

$$\frac{-i\omega}{2} [\tilde{A}(k) - \tilde{A}^*(-k)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \dot{f}(z, 0) e^{-ikz} dz$$

$$\frac{1}{2} [\tilde{A}(k) - \tilde{A}^*(-k)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{i}{\omega} \dot{f}(z, 0) \right] e^{-ikz} dz \quad \text{----- (ii)}$$

Adding equation (i) & (ii) we get

$$\tilde{A}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[f(z, 0) + \frac{i}{\omega} \dot{f}(z, 0) \right] e^{-ikz} dz$$

Example 4:

Show that the standing wave $f(z, t) = A \sin(kz) \cos(kvt)$ satisfies the wave equation, and express it as the sum of a wave traveling to the left and a wave traveling to the right .

Solution :

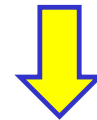
Given , $f(z, t) = A \sin(kz) \cos(kvt)$ ----- (a)

$$\frac{\partial f}{\partial z} = Ak \cos(kz) \cos(kvt)$$

$$\frac{\partial^2 f}{\partial z^2} = -Ak^2 \sin(kz) \cos(kvt)$$

$$\frac{\partial f}{\partial t} = -A(vk) \sin(kz) \sin(kvt)$$

$$\frac{\partial^2 f}{\partial t^2} = -A(vk)^2 \sin(kz) \cos(kvt)$$



$$\frac{\partial^2 f}{\partial t^2} = v^2 \frac{\partial^2 f}{\partial z^2}$$

Therefore equation (a) satisfy the wave equation .

$$f(z, t) = A \sin(kz) \cos(kvt)$$

$$f(z, t) = \frac{A}{2} [2 \sin(kz) \cos(kvt)]$$

$2 \sin(A) \cos(B) = \sin(A - B) + \sin(A + B)$

$$f(z, t) = \frac{A}{2} [\sin(kz - kvt) + \sin(kz + kvt)] \quad \text{----- (b)}$$

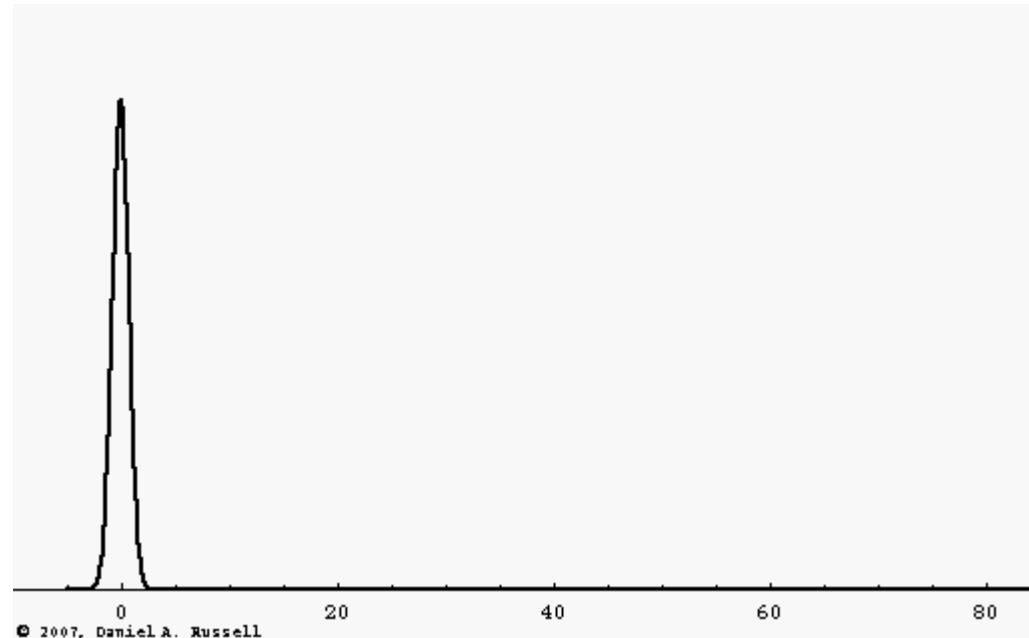
General solution of wave equation is

$$f(z, t) = g(z - vt) + h(z + vt)$$

$$\begin{aligned} \Rightarrow g(z - vt) &= \frac{A}{2} \sin k(z - vt) \\ \Rightarrow h(z + vt) &= \frac{A}{2} \cos k(z + vt) \end{aligned} \quad \left. \vphantom{\begin{aligned} \Rightarrow g(z - vt) &= \frac{A}{2} \sin k(z - vt) \\ \Rightarrow h(z + vt) &= \frac{A}{2} \cos k(z + vt) \end{aligned}} \right\} \text{On comparing with eq. (b)}$$

So equation (a) can be expressed as the sum of a wave traveling to the left and a wave traveling to the right .

Dispersion causes the shape of a wave pulse to change as it travels



If the wave speed depends **only** on the physical properties of the medium (i.e., the elastic and inertia properties of a mechanical medium, or the relative permeability and permittivity for EM waves) then the wave speed is a constant, independent of frequency. Such a medium is called a nondispersive medium and waves traveling through this medium will maintain a constant shape.

This is illustrated by the **black wave pulse** in the animation at left. The pulse is a Gaussian function, containing multiple frequencies. Since the wave speed is constant, all frequencies travel at the same speed and the pulse maintains constant shape.

There are many examples of dispersive media where, for various reasons, the wave speed depends on the frequency of the wave. The blue wave pulse in the animation is the same Gaussian function as the black pulse, consisting of a large number of frequency components added together. However, now the wave speed depends on frequency, with higher frequencies traveling faster than lower frequencies. As a result, the wave pulse spreads out and changes shape as it travels.

Classical Systems – Interference of Wave Amplitudes

Consider two traveling waves interfering with each other in a non-dispersive medium with different frequencies and amplitudes.

For a non-dispersive medium, this means that

$$v = f_1 \lambda_1 = f_2 \lambda_2 = \frac{\omega_1}{k_1} = \frac{\omega_2}{k_2}$$

$$\omega_1 = 2\pi f_1$$

$$\omega_2 = 2\pi f_2$$

$$f_{total}(z, t) = \tilde{f}_1(z, t) + \tilde{f}_2(z, t)$$

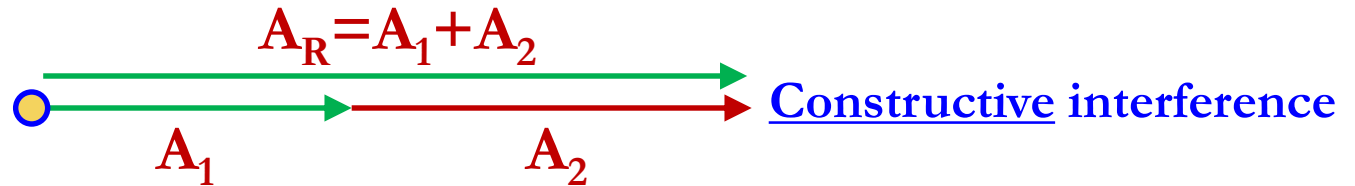
$$f_{total}(z, t) = A_1 e^{i(k_1 z - \omega_1 t + \delta_1)} + A_2 e^{i(k_2 z - \omega_2 t + \delta_2)}$$

Easy cases of phase relations between the two waves

Case-I

$$\delta_1 = \delta_2 \quad \text{In phase}$$

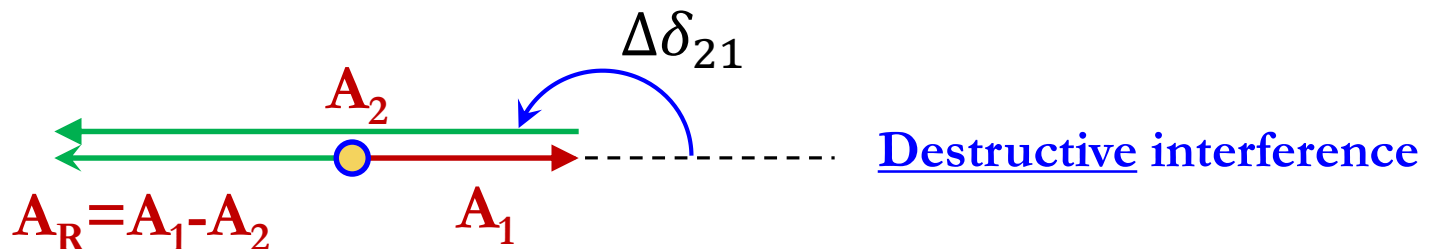
$$\Delta\delta_{21} = \delta_2 - \delta_1 = 0$$



Case-II

$$\delta_2 = \delta_1 + \pi \text{ (180}^\circ \text{ out of phase)}$$

$$\Delta\delta_{21} = \delta_2 - \delta_1 = \pi \text{ radians} = 180^\circ$$



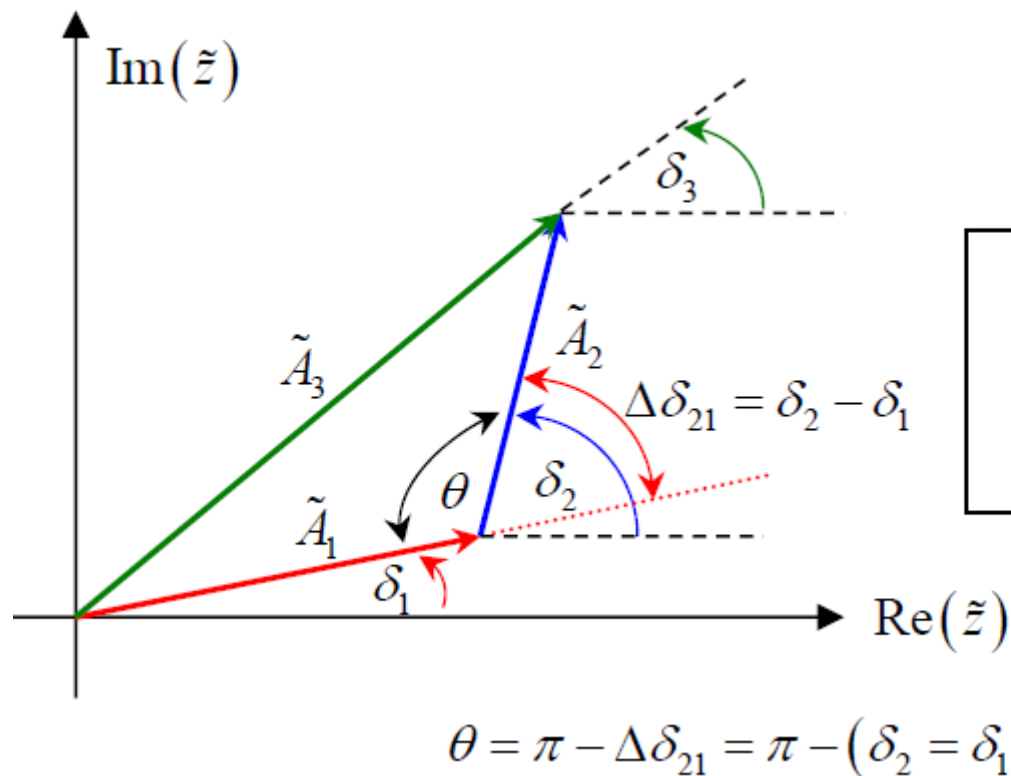
The General Case:

$$\tilde{f}_1(z, t) \equiv \tilde{A}_1 e^{i(kz - \omega t)}$$

$$\tilde{f}_2(z, t) \equiv \tilde{A}_2 e^{i(kz - \omega t)}$$

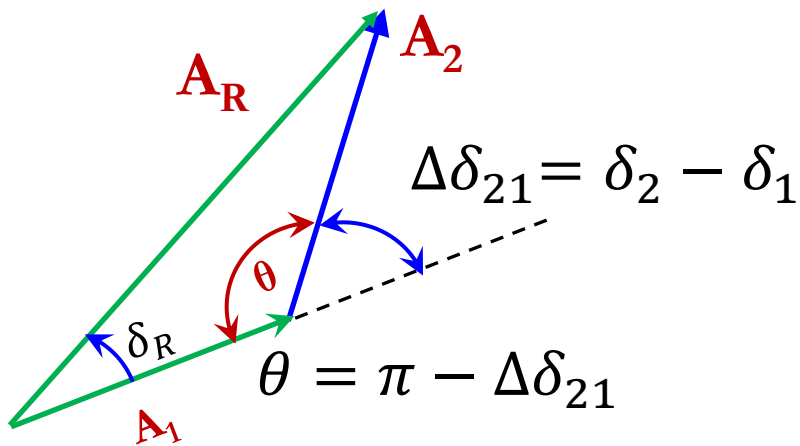
$$\tilde{f}_3(z, t) = \tilde{f}_1(z, t) + \tilde{f}_2(z, t)$$

$$\tilde{f}_3(z, t) \equiv \tilde{A}_3 e^{i(kz - \omega t)}$$



n.b. phases are
always defined
relative to the
Real axis

An **Argand diagram** is a plot of complex numbers as points in the complex plane using the x-axis as the real axis and y-axis as the imaginary axis.



$$|\tilde{A}_3| = \sqrt{\tilde{A}_3 \cdot \tilde{A}_3^*} = \sqrt{(\tilde{A}_1 + \tilde{A}_2) \cdot (\tilde{A}_1 + \tilde{A}_2)^*}$$

$$= \sqrt{(A_1 e^{i\delta_1} + A_2 e^{i\delta_2})(A_1 e^{-i\delta_1} + A_2 e^{-i\delta_2})}$$

$$= \sqrt{A_1^2 + A_2^2 + A_1 A_2 (e^{i\delta_1} e^{-i\delta_2} + e^{-i\delta_1} e^{i\delta_2})}$$

$$= \sqrt{A_1^2 + A_2^2 + A_1 A_2 [2 \cos(\delta_2 - \delta_1)]}$$

From the diagram, we see that the magnitude of the resultant (*i.e.* net or total) amplitude A_R is:

$$A_R = A_3 = |\tilde{f}_{total}(z, t)| = \sqrt{A_1^2 + A_2^2 - 2A_1 A_2 \cos \theta}$$

$$= \sqrt{A_1^2 + A_2^2 - 2A_1 A_2 \cos(\pi - \Delta\delta_{21})}$$

$$\cos(\pi - \Delta\delta_{21}) = \cos \pi \cos \Delta\delta_{21} + \sin \pi \sin \Delta\delta_{21}$$

$$\cos(\pi - \Delta\delta_{21}) = \cos \pi \cos \Delta\delta_{21} + \sin \pi \sin \Delta\delta_{21}$$

$$\cos(\pi - \Delta\delta_{21}) = \cos \pi \cos \Delta\delta_{21} + \sin \pi \sin \Delta\delta_{21}$$

$$\cos(\pi - \Delta\delta_{21}) = -\cos \Delta\delta_{21}$$

$$A_R = A_3 = |\tilde{f}_{total}(z, t)| = \sqrt{A_1^2 + A_2^2 - 2A_1A_2 \cos(\pi - \Delta\delta_{21})}$$

$$= \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta\delta_{21}}$$

$$= \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\delta_2 - \delta_1)}$$

$$A_R = |\tilde{f}_{total}(z, t)| = \sqrt{\tilde{f}_{total}(z, t) \times \underbrace{\tilde{f}_{total}^*(z, t)}_{\text{Complex conjugate of } \tilde{f}_{total}(z, t)}}$$

Complex conjugate of $\tilde{f}_{total}(z, t)$

Thus if:

$$\tilde{f}_1(z, t) = A_1 e^{i(kz - \omega t + \delta_1)}$$

$$\tilde{f}_2(z, t) = A_2 e^{i(kz - \omega t + \delta_2)}$$

$$\tilde{f}_{total}(z, t) = A_1 e^{i(kz - \omega t + \delta_1)} + A_2 e^{i(kz - \omega t + \delta_2)}$$

$$\tilde{f}_{total}(z, t) = A_R e^{i(kz - \omega t + \delta_R)}$$

$$A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\delta_2 - \delta_1)} = \sqrt{A_1^2 + A_2^2 + \underbrace{2A_1A_2 \cos(\delta_1 - \delta_2)}}_{\text{Since cosine is an even function}}$$

Since cosine is
an even
function

Then if the two waves have equal amplitudes

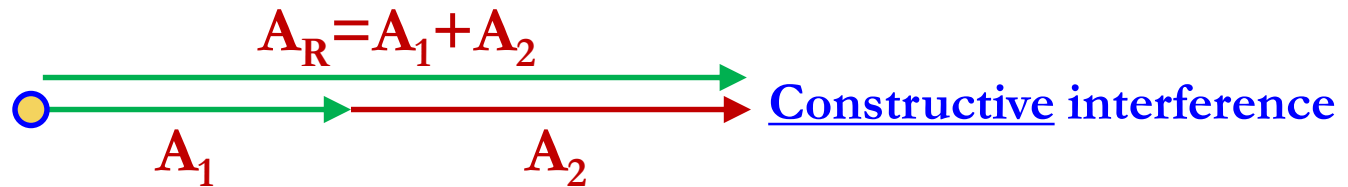
$$A_1 = A_2$$

Case-I

If $\delta_1 = \delta_2$ (in phase with each other), then $\Delta\delta = 0$ and hence:

$$\cos(\delta_2 - \delta_1) = 1 \quad \text{Total constructive interference}$$

$$A_R = \sqrt{A^2 + A^2 + 2A^2} = 2A \quad \text{Resultant Amplitude}$$

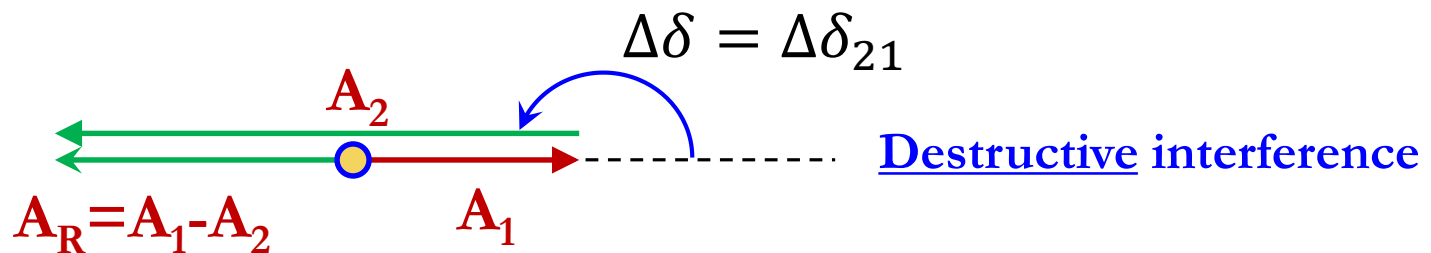


Case-II

If $\delta_2 = \delta_1 \pm \pi$, then $\Delta\delta = \pm\pi$ (out of phase with each other) hence:

$$\cos(\delta_2 - \delta_1) = -1 \quad \text{Total destructive interference}$$

$$A_R = \sqrt{A^2 + A^2 - 2A^2} = 0 \quad \text{Resultant Amplitude}$$



$$A_R = 0 \text{ if } (A_1 = A_2)$$

Note that classical wave interference effects can occur at the amplitude level even if $f_1 \neq f_2$ (Different frequency f still you can see the Interference)

Examples:

Sound waves on strings, in air,...

Electronic signals

EM waves

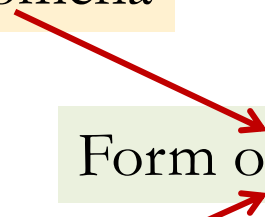
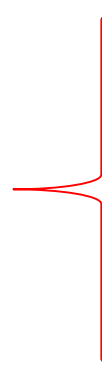
$$f_1 \approx f_2$$

Beats phenomena

$$f_1 \gg f_2$$

Modulation phenomena

Form of interference



Wave Intensity, (**I**)

Wave intensities are proportional to the square of Amplitude

$$I = \tilde{A}\tilde{A}^*$$

The total intensity

$$I \propto \tilde{f}_{total}(z, t) \cdot \tilde{f}_{total}^*(z, t)$$

In above previous example, the individual normalized intensities are

$$I_1 = A_1^2$$

&

$$I_2 = A_2^2$$

The corresponding resultant normalized intensity is:

$$A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\delta_2 - \delta_1)}$$

$$I_R = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos(\delta_2 - \delta_1)$$

Then for the special case of $A_1 = A_2 = A$ or equivalently $I_1 = I_2 = I_0$,
Then *e.g.* for ($z = 0, t = 0$):

Case-I

If $\delta_1 = \delta_2$ (in phase with each other), then $\Delta\delta = 0$ and hence:

$$I_R = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos(\delta_2 - \delta_1)$$

$$\cos(\delta_2 - \delta_1) = +1 \quad \Rightarrow \quad I_R = 4I_0$$

Constructive Interference

Case-II

If $\delta_2 = \delta_1 \pm \pi$, then $\Delta\delta = \pm\pi$ (out of phase with each other) hence:

$$\cos(\delta_2 - \delta_1) = -1 \quad \Rightarrow \quad I_R = 0$$

Total destructive interference

Standing Waves

A standing wave (*stationary* in space) is formed by superposing two identical traveling waves, except that one is left-going traveling wave and the other is right-going:

$$f_{total}(z, t) = g(z - vt) + g(z + vt)$$

Same wave going Left and Right

Consider the linear superposition of a left-going and a right-going traveling wave

$$g(z - vt) = A \sin(kz - \omega t)$$

$$g(z + vt) = A \sin(kz + \omega t)$$

$$f_{total}(z, t) = A \sin(kz - \omega t) + A \sin(kz + \omega t)$$

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$f_{total}(z, t) = 2A \sin kz \cos \omega t$$

$$f_{total}(z, t) = A(\sin kz \cos \omega t + \cos kz \sin \omega t) + A(\sin kz \cos \omega t - \cos kz \sin \omega t)$$

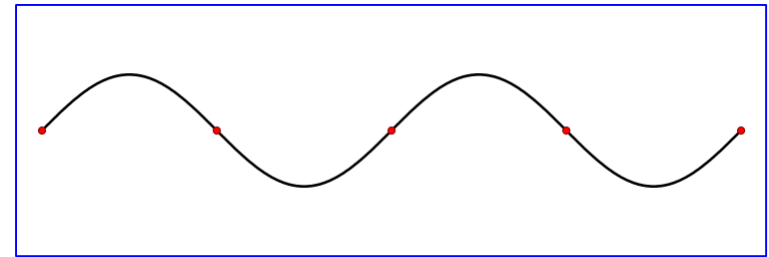
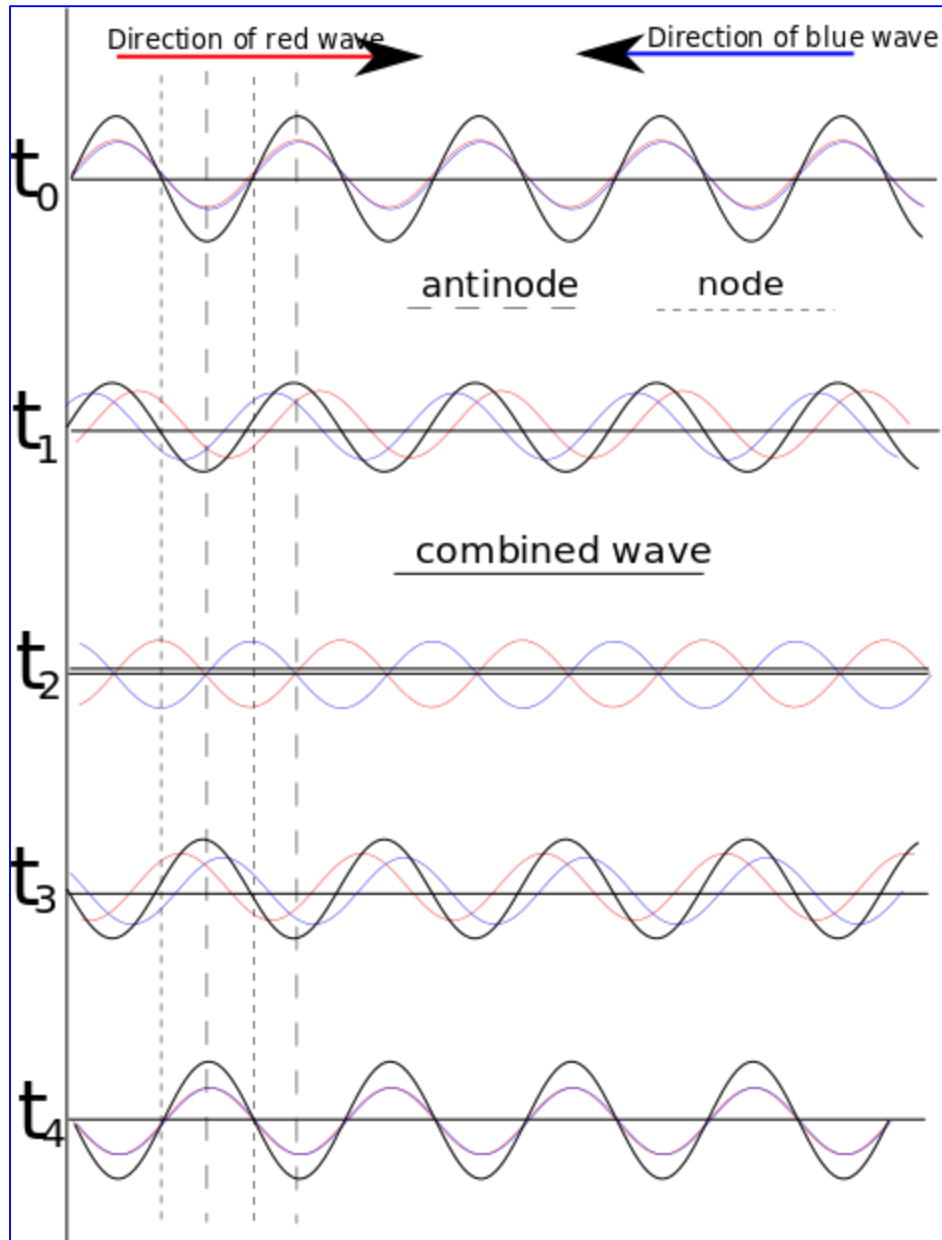
$$f_{total}(z, t) = A(\sin kz \cos \omega t + \cancel{\cos kz} \sin \omega t) + A(\sin kz \cos \omega t - \cancel{\cos kz} \sin \omega t)$$

$$f_{total}(z, t) = 2A \sin kz \cos \omega t$$

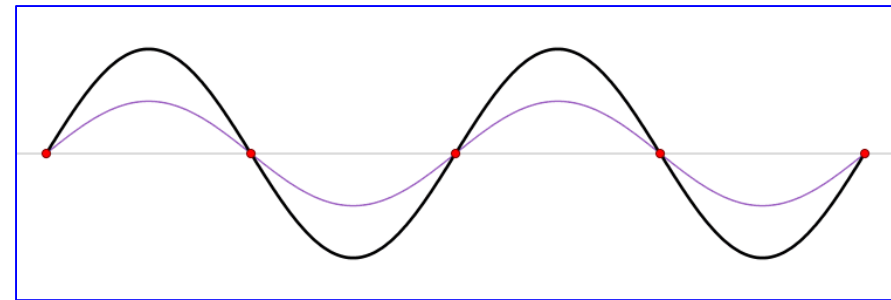
1-D standing wave (*i.e.* does not propagate/move in longitudinal $\pm \hat{z}$ -direction)

This describes a wave that oscillates in time, but has a spatial dependence that is stationary: **$\sin kz$**

Two opposing waves combine to form a standing wave.



Standing wave in stationary medium. The red dots represent the wave nodes.



A standing wave (black) depicted as the sum of two propagating waves traveling in opposite directions (red and blue).

Standing first discovered by Franz Melde, who coined the term "standing wave" around 1860

$$f_{total}(z, t) = 2A \sin kz \cos \omega t$$

At locations $x = 0, \lambda/2, \lambda, 3\lambda/2, \dots$ called the nodes the amplitude is always zero, whereas at locations $x = \lambda/4, 3\lambda/4, 5\lambda/4, \dots$ called the anti-nodes, the amplitude is maximum. The distance between two conjugative nodes or anti-nodes is $\lambda/2$.

EM Waves inside the Matter.

Electromagnetic Wave Propagation in Linear Media

We now consider EM wave propagation inside **linear matter**, but **only** in regions where there are **NO free charges** ($\sigma_{\text{free}} = 0$) and/or **free currents** ($\vec{K}_{\text{free}} = 0$) (i.e. the **linear** medium is an **insulator** / a **non-conductor**).

For this situation, Maxwell's equations (in differential form) become:

$$\nabla \cdot \vec{D} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$\vec{D}(r, t) = \epsilon \vec{E}(r, t)$$

$$\vec{H}(r, t) = \frac{1}{\mu} \vec{B}(r, t)$$

$\varepsilon = \text{Electric permittivity of the medium}$

$\mu = \text{Magnetic permittivity of the medium}$

$\varepsilon = \varepsilon_0(1 + \chi_e), \quad \chi_e = \text{Electric Susceptibility of the medium}$

$\mu = \mu_0(1 + \chi_m), \quad \chi_m = \text{Magnetic permeability of the medium}$

$\varepsilon_0 = \text{Electrical permittivity of free space} = 8.85 \times 10^{-12} \text{ Farads/m}$

$\mu_0 = \text{Magnetic permittivity of free space} = 4\pi \times 10^{-7} \text{ Henrys/m}$

Thus, Maxwell's equations for the $\vec{E}$ and $\vec{B}$ fields inside the linear, homogeneous and isotropic non-conducting medium become:

$$\nabla \cdot \vec{E}(r, t) = 0$$

$$\nabla \times \vec{E}(r, t) = -\frac{\partial \vec{B}(r, t)}{\partial t}$$

$$\nabla \cdot \vec{B}(r, t) = 0$$

$$\nabla \times \vec{B}(r, t) = \mu\varepsilon \frac{\partial \vec{E}(r, t)}{\partial t}$$

$$\nabla \cdot \vec{E}(r, t) = 0$$

$$\nabla \times \vec{E}(r, t) = -\frac{\partial \vec{B}(r, t)}{\partial t}$$

$$\nabla \cdot \vec{B}(r, t) = 0$$

$$\nabla \times \vec{B}(r, t) = \mu\epsilon \frac{\partial \vec{E}(r, t)}{\partial t}$$

Note that the above four relations are (almost) identical to those for *EM* waves in free space.

We simply replace the macroscopic *EM* parameters associated with the vacuum with $\{\epsilon_0, \mu_0\}$ those associated with the linear, homogeneous and isotropic medium $\{\epsilon, \mu\}$.

In free space/vacuum, the speed of propagation of *EM* waves is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \frac{\text{m}}{\text{s}}$$

$\vec{E}$ and $\vec{B}$ fields in vacuum obey the wave equation

$$\nabla^2 \vec{E} = \epsilon_0 \mu_0 \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \epsilon_0 \mu_0 \frac{\partial^2 \vec{B}}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$

In a linear/homogeneous/isotropic medium, the speed of propagation of *EM* waves is:

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

The **E** and **B** fields in the medium obey the following wave equation:

$$\nabla^2 \vec{E} = \epsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \epsilon\mu \frac{\partial^2 \vec{B}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$

For linear/homogeneous/isotropic media

$$\varepsilon = K_e \varepsilon_0 = (1 + \chi_e) \varepsilon_0 \quad K_e = \frac{\varepsilon}{\varepsilon_0} \quad \text{Relative electric permittivity}$$

$$\mu = K_m \mu_0 = (1 + \chi_m) \mu_0 \quad K_m = \frac{\mu}{\mu_0} \quad \text{Relative magnetic permeability}$$

$$v = \frac{1}{\sqrt{\mu \varepsilon}}$$

$$v = \frac{1}{\sqrt{K_e \varepsilon_0 K_m \mu_0}} = \frac{1}{\sqrt{K_e K_m}} \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \frac{c}{\sqrt{K_e K_m}}$$

Now if:

$$K_e = \frac{\varepsilon}{\varepsilon_0} = (1 + \chi_e) \geq 1 \quad K_m = \frac{\mu}{\mu_0} = (1 + \chi_m) \geq 1$$

$$K_e K_m \geq 1$$

True for a wide variety of common/everyday materials – gases, liquids & solids

$$\sqrt{K_e K_m} \geq 1$$



$$\frac{1}{\sqrt{K_e K_m}} \leq 1$$

$$v = \frac{c}{\sqrt{K_e K_m}} \leq c$$

We can now define the index of refraction {*n.b.* a dimensionless quantity} of the linear/homogeneous/isotropic medium as:

$$n = \sqrt{K_e K_m} = \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}}$$

Thus, for linear/homogeneous/isotropic media:

$$v = \frac{c}{n} \leq c$$

$$n \geq 1 \text{ (usually)}$$

Usually ε and μ are not constant but they are frequency dependent

$$\varepsilon = \varepsilon(\omega)$$

$$\mu = \mu(\omega)$$

$$\omega = 2\pi f$$

$$K_e = K_e(\omega) = \frac{\varepsilon(\omega)}{\varepsilon_0} = (1 + \chi_e(\omega))$$

$$K_m = K_m(\omega) = \frac{\mu(\omega)}{\mu_0} = (1 + \chi_m(\omega))$$

$$n = n(\omega) = \sqrt{K_e(\omega)K_m(\omega)} = \sqrt{\frac{\varepsilon(\omega)\mu(\omega)}{\varepsilon_0\mu_0}} = \sqrt{(1 + \chi_e(\omega))(1 + \chi_m(\omega))}$$

$$v(\omega) = \frac{c}{n(\omega)}$$

For now, we will ignore frequency-dependent effects, for simplicity, *i.e.*

$$v(\omega) = \frac{c}{n} = \textit{Constant}$$

$$n = \sqrt{(1 + \chi_e)(1 + \chi_m)} = \textit{Constant}$$

Now for many (but not all) linear/homogeneous/isotropic materials

$$\mu = (1 + \chi_m)\mu_0 \approx \mu_0$$

This is true for many paramagnetic and diamagnetic-type materials since $\chi_m \sim 10^{-8} \sim 0$

$$K_m = \frac{\mu}{\mu_0} = 1$$

$$n = \sqrt{K_e K_m} = \sqrt{K_e}$$

$$v = \frac{c}{n} = \frac{c}{\sqrt{K_e}}$$

The instantaneous EM energy density associated with a linear/homogeneous/isotropic material:

$$u_{em} = (u_e + u_m)$$

$$u_{em} = (u_e + u_m)$$

$$u_{em} = \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H})$$

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{B^2}{\mu} \right)$$

Joules/m³

$$\vec{D} = \epsilon \vec{E} \quad \vec{H} = \frac{1}{\mu} \vec{B}$$

But

$$\vec{B} = \frac{1}{v} \hat{k} \times \vec{E} \quad \& \quad v^2 = \frac{1}{\epsilon \mu}$$

$$v^2 = \frac{1}{\mu\epsilon}$$

$$\vec{B} = \frac{1}{v} \hat{k} \times \vec{E} \quad \& \quad v^2 = \frac{1}{\epsilon\mu}$$

Hence

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{E^2}{\mu v^2} \right)$$

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{B^2}{\mu} \right)$$

$$u_{em} = \frac{1}{2} \left(\epsilon E^2 + \frac{\mu \epsilon E^2}{\mu} \right)$$

$$u_{em} = \frac{1}{2} (\epsilon E^2 + \epsilon E^2)$$

$$u_{em} = \epsilon E^2$$

Hence for linear media

$$u_{em} = \frac{1}{2} (\epsilon E^2 + \epsilon E^2)$$

$$u_e = u_m$$

The instantaneous Poynting's vector associated with a linear/homogeneous/isotropic material:

$$\vec{S} = \vec{E} \times \vec{H} = \frac{1}{\mu} (\vec{E} \times \vec{B})$$

$$\vec{B} = \frac{1}{v} \hat{k} \times \vec{E}$$

$$\vec{S} = \frac{1}{\mu} (\vec{E} \times \vec{B}) = \frac{1}{\mu v} E^2 \hat{k}$$

$$v^2 = \frac{1}{\mu \epsilon}$$

$$\vec{S} = \frac{v}{\mu v^2} E^2 \hat{k}$$

$$\vec{S} = \frac{v}{\mu v^2} E^2 \hat{k}$$

$$v^2 = \frac{1}{\mu \epsilon}$$

$$\vec{S} = v \epsilon E^2 \hat{k}$$

$$u_{em} = \epsilon E^2$$

$$\vec{S} = v u_{em} \hat{k}$$

Lets define the Vector impedance associated with a linear/homogeneous/isotropic material is given by the relation:

$$\vec{Z} = \vec{E} \times \vec{H}^{-1}$$

$$\vec{Z} = \frac{(\vec{E} \times \vec{H})}{|\vec{H}|^2}$$

$$\vec{Z} = \frac{(\vec{E} \times \vec{H})}{|\vec{H}|^2}$$

$$\vec{Z} = \mu v \hat{k}$$

Vector impedance

$$\vec{S} = v u_{em} \hat{k}$$

$$u_{em} = \varepsilon E^2$$

$$\vec{S} = v \varepsilon E^2 \hat{k}$$

$$\vec{Z} = \sqrt{\frac{\mu}{\varepsilon}} \hat{k}$$

Ohms

For monochromatic (*i.e.* sinusoidal, single frequency) plane *EM* waves propagating in a linear/homogeneous/isotropic medium, **E** and **B** satisfy/obey the wave equation:

$$\nabla^2 \vec{E} = \varepsilon \mu \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \varepsilon \mu \frac{\partial^2 \vec{B}}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$